

PROGRAM:

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#include<stdio.h>

// Function to swap two elements in the array

void swap(int* a, int* b) {

    int temp = *a;

    *a = *b;

    *b = temp; }

// Function to partition the array and return the index of the pivot

int partition(int arr[], int low, int high) {

    int pivot = arr[high];

    // Choosing the last element as the pivot

    int i = low - 1;

    // Index of smaller element

    for (int j = low; j < high; j++) {

        // If current element is smaller than the pivot

        if (arr[j] < pivot) {

            i++;

            swap(&arr[i], &arr[j]);

        }

    }

    // Place the pivot element in its correct position

    swap(&arr[i + 1], &arr[high]);

    return i + 1;

}

// Function to implement Quick Sort

void quickSort(int arr[], int low, int high) {

    if (low < high) {

        // Partitioning index
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        int pi = partition(arr, low, high);

        // Recursive sorting of the sub-arrays

        quickSort(arr, low, pi - 1);

        quickSort(arr, pi + 1, high);

    }

}

// Function to print the array

void printArray(int arr[], int size) {

    for (int i = 0; i < size; i++)

        printf("%d ", arr[i]);    printf("\n");

}

int main() {

    int arr[] = {12, 7, 11, 13, 5, 6};

    int n = sizeof(arr) / sizeof(arr[0]);

    printf("Original array:\n");

    printArray(arr, n);

    quickSort(arr, 0, n - 1);

    printf("Sorted array:\n");

    printArray(arr, n);

    return 0;

}
```

OUTPUT:

Original array: 12 7 11 13 5 6

Sorted array: 5 6 7 11 12 13